

$$41. \text{ Given exp. } = \left(\frac{41}{6} \times \frac{16}{3} + \frac{53}{3} \times \frac{9}{2} \right) = \left(\frac{328}{9} + \frac{159}{2} \right) \\ = \frac{656 + 1431}{18} = \frac{2087}{18} = 115 \frac{17}{18}.$$

$$42. \text{ Given exp. } = \frac{24}{5} \div \frac{32}{5} = \frac{24}{5} \times \frac{5}{32} = \frac{3}{4}.$$

$$43. \text{ Given exp. } = \frac{5}{4} + \frac{14}{9} \times \frac{13}{8} \times \frac{2}{13} = \frac{5}{4} + \frac{7}{18} = \frac{45 + 14}{36} = \frac{59}{36} = 1 \frac{23}{36}.$$

$$44. \text{ Given exp. } = \frac{225}{836} \times \frac{152}{245} \div \frac{120}{77} = \frac{225}{836} \times \frac{152}{245} \times \frac{77}{120} = \frac{3}{28}.$$

$$45. \text{ Given exp. } = \left(\frac{82}{5} - \frac{181}{15} \right) \div \frac{247}{81} = \left(\frac{246 - 181}{15} \right) \times \frac{81}{247} \\ = \frac{65}{15} \times \frac{81}{247} = \frac{27}{19} = 1 \frac{8}{19}.$$

$$46. \text{ Let } 18 \frac{3}{4} \times x \div \frac{6}{37} = 1480.$$

$$\text{Then, } \frac{75}{4} \times x \times \frac{37}{6} = 1480 \Leftrightarrow x = \frac{1480 \times 4 \times 6}{75 \times 37} = \frac{64}{5} = 12 \frac{4}{5}.$$

$$47. \text{ Given exp. } = \frac{3}{2} \times \frac{11}{5} \div \frac{5}{4} \div \frac{33}{15} = \frac{3}{2} \times \frac{11}{5} \times \frac{4}{5} \div \frac{33}{15} \\ = \frac{3}{2} \times \frac{11}{5} \times \frac{4}{5} \times \frac{15}{33} = \frac{6}{5} = 1 \frac{1}{5}.$$

$$48. \quad 3 \frac{x}{7} \times 2 \frac{y}{5} = 10 \Leftrightarrow \frac{21+x}{7} \times \frac{10+y}{5} = 10 \\ \Leftrightarrow (21+x)(10+y) = 350 \quad \dots(i) \\ \text{Clearly, } x = 4, y = 4 \text{ satisfy the equation (i).}$$

$$49. \text{ Given exp. } = 1 + 2 \div \left\{ 1 + 2 \div \frac{4}{3} \right\} = 1 + 2 \div \left\{ 1 + 2 \times \frac{3}{4} \right\} \\ = 1 + 2 \div \left\{ 1 + \frac{3}{2} \right\} = 1 + 2 \div \frac{5}{2} = 1 + 2 \times \frac{2}{5} \\ = 1 + \frac{4}{5} = \frac{9}{5} = 1 \frac{4}{5}.$$

$$50. \text{ Given exp. } = \frac{27}{8} \text{ of } \frac{3}{15} \div \frac{5}{36} \text{ of } \frac{5}{49} = \frac{81}{10} \div \frac{25}{63} \\ = \frac{81}{10} \times \frac{63}{25} = \frac{5103}{250} = 20 \frac{103}{250}.$$

$$51. \text{ Given exp. } = \left(-\frac{2}{3} - \frac{1}{3} \right) + \left(\frac{4}{5} + \frac{1}{5} \right) + \left(\frac{3}{4} - \frac{1}{2} \right) \\ = \left(\frac{2}{3} - \frac{4}{3} + \frac{1}{3} \right) - \left(\frac{1}{5} + \frac{4}{5} \right) + \frac{1}{2} \\ = \frac{-1+1+\frac{1}{4}}{-\frac{1}{3}-1+\frac{1}{2}} = \frac{\frac{1}{4}}{\frac{-2-6+3}{6}} = \frac{\frac{1}{4}}{-\frac{5}{6}} \\ = \frac{1}{4} \times \left(-\frac{6}{5} \right) = -\frac{3}{10}.$$

$$52. \quad \left(\frac{\frac{1}{2} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6}}{\frac{2}{5} - \frac{5}{9} + \frac{3}{5} - \frac{7}{18}} \right) = \left(\frac{\frac{30-15+12-10}{60}}{\frac{2}{5} + \frac{3}{5} - \left(\frac{5}{9} + \frac{7}{18} \right)} \right)$$

$$= \left(\frac{\frac{17}{60}}{1 - \frac{17}{18}} \right) = \left(\frac{17}{60} \times 18 \right) = \frac{51}{10} = 5 \frac{1}{10}.$$

$$53. \quad \left(34 \times 4 \frac{1}{2} \right) = 34 \times \left(4 + \frac{1}{2} \right) = (34 \times 4) + \left(34 \times \frac{1}{2} \right) \\ = (30+4) \times 4 + \left(34 \times \frac{1}{2} \right) \\ = (30 \times 4) + (4 \times 4) + \left(34 \times \frac{1}{2} \right).$$

$$54. \text{ Let } \frac{3}{8} \text{ of } 168 \times 15 \div 5 + x = 549 \div 9 + 235. \\ \text{Then, } 63 \times 15 \div 5 + x = 61 + 235 \Leftrightarrow 63 \times 3 + x = 296 \\ \Leftrightarrow 189 + x = 296 \\ \Leftrightarrow x = 107.$$

$$55. \text{ Let } \frac{5}{3} \div \frac{2}{7} \times \frac{x}{7} = \frac{5}{4} \times \frac{2}{3} \div \frac{1}{6}.$$

$$\text{Then, } \frac{5}{3} \times \frac{7}{2} \times \frac{x}{7} = \frac{5}{4} \times \frac{2}{3} \times 6$$

$$\Leftrightarrow \frac{5}{6} x = 5 \Leftrightarrow x = \left(\frac{5 \times 6}{5} \right) = 6.$$

$$56. \text{ Let } 5 \frac{2}{3} \div x \frac{5}{6} = 2.$$

$$\text{Then, } \frac{17}{3} \div x \frac{5}{6} = 2 \Leftrightarrow x \frac{5}{6} = \frac{17}{3} \times \frac{1}{2} = \frac{17}{6} \Leftrightarrow x \frac{5}{6} = 2 \frac{5}{6}.$$

$$\therefore x = 2.$$

$$57. \text{ Given equation is: } \frac{(5x+1)}{x} \times \frac{(4y+3)}{4} = 20 \\ \Leftrightarrow (5x+1)(4y+3) = 80x \quad \dots(i) \\ \text{Clearly, } x = 3 \text{ and } y = 3 \text{ satisfy (i).}$$

$$58. \text{ Required difference } = \frac{19}{16} - \frac{16}{19} = \frac{19^2 - 16^2}{304} \\ = \frac{(19+16)(19-16)}{304} = \frac{35 \times 3}{304} = \frac{105}{304}.$$

$$59. \text{ Required number } = \frac{37 \frac{1}{2}}{1/8} = \frac{75/2}{1/8} = \frac{75}{2} \times 8 = 300.$$

$$60. \quad a = (4 \div 3) \div 3 \div 4 = \frac{4}{3} \times \frac{1}{3} \times \frac{1}{4} = \frac{1}{9}.$$

$$b = 4 \div (3 \div 3) \div 4 = 4 \div 1 \div 4 = 1.$$

$$c = 4 \div 3 \div (3 \div 4) = 4 \div 3 \div \frac{3}{4} = \frac{4}{3} \times \frac{4}{3} = \frac{16}{9}.$$

Clearly, c is the greatest.

$$61. \quad \text{I. } \frac{3}{4} \div \frac{5}{6} = \frac{3}{4} \times \frac{6}{5} = \frac{9}{10}.$$

$$\text{II. } 3 \div [(4 \div 5) \div 6] = 3 \div \left(\frac{4}{5} \times \frac{1}{6} \right) = 3 \div \frac{4}{30} = 3 \times \frac{30}{4} = \frac{45}{2}.$$

$$\text{III. } [3 \div (4 \div 5)] \div 6 = \left[3 \div \frac{4}{5} \right] \div 6 = \left[3 \times \frac{5}{4} \right] \div 6 = \frac{15}{4} \times \frac{1}{6} = \frac{5}{8}.$$

$$\text{IV. } 3 \div 4 \div (5 \div 6) = 3 \div 4 \div \frac{5}{6} = \frac{3}{4} \times \frac{6}{5} = \frac{9}{10}.$$

So, I and IV are equal.